# Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Area and perimeter**

A rectangle has the dimensions shown below. Find its area and perimeter.

**2** **[Foundational]** • **Target: Area and perimeter**

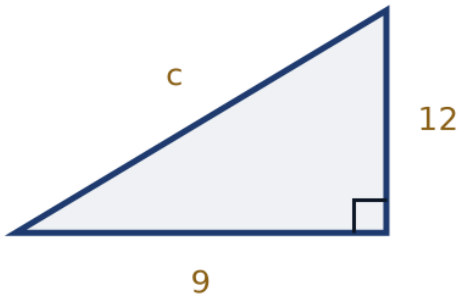
Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

**3** **[Foundational]** • **Target: Area and perimeter**

A triangle has base 15 and height 10. Find its area.

**4** **[Foundational]** • **Target: The Pythagorean theorem**

A right triangle has the two legs shown below. Find the length of the hypotenuse $c$.




**5** **[Foundational]** • **Target: The Pythagorean theorem**

A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

**6** **[Foundational]** • **Target: Volume and surface area**

Find the volume of a cylinder with radius 4 and height 9, in terms of $\pi$.

**7** **[Foundational]** • **Target: Volume and surface area**

Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Similar triangles and proportional reasoning**

Triangle $ABC$ is similar to triangle $DEF$, with $AB = 8$, $DE = 12$, and $BC = 10$. Find $EF$.

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**9** [Intermediate] • Target: Similar triangles and proportional reasoning

A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

**10** [Intermediate] • Target: Angle relationships

Two parallel lines are cut by a transversal. If one angle measures $72^\circ$, find its corresponding angle and its co-interior (same-side interior) angle.

**11** [Intermediate] • Target: Angle relationships

The three angles of a triangle are in the ratio $2:3:4$. Find each angle.

**12** [Intermediate] • Target: Angle relationships

Two angles are complementary, and one is $34^\circ$ more than the other. Find both angles.

**13** [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the distance between the points $(3, -2)$ and $(9, 6)$.

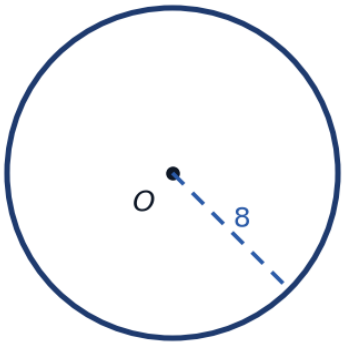
**14** [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A circle has the radius shown below. Find its area and circumference, in terms of $\pi$.



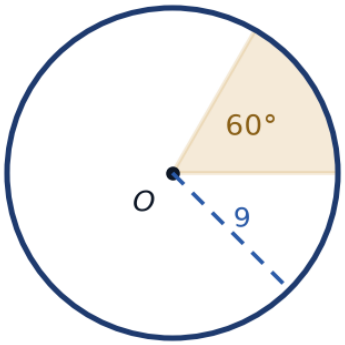

**16** [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 8.01 Exponentials & Logarithms

Write the equation of a circle with center $(3, -4)$ and radius 6.

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**17** [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 1.01 Heart of Algebra

A circle has the radius and shaded sector shown below. Find the area of the shaded $60^\circ$ sector, in terms of $\pi$.




**18** [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 2.01 Quadratics & Polynomials

A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of $\pi$.

**19** [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 3.01 Functions & Graphs

A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

**20** [Challenging] • Target: A well-designed word problem: designing a park • Cross-Lesson: 4.01 Ratios, Rates & Data

This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of $\pi$. (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of $\pi$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Area and perimeter

A rectangle has length 23 and width 5. Find its area and perimeter.

**a.** $115, 56$

**b.** $56, 115$

**c.** $119, 56$

**d.** $115, 54$

**22** [Foundational] • Target: The Pythagorean theorem

A right triangle has legs 3 and 4. Find the length of the hypotenuse $c$.

**a.** $5$

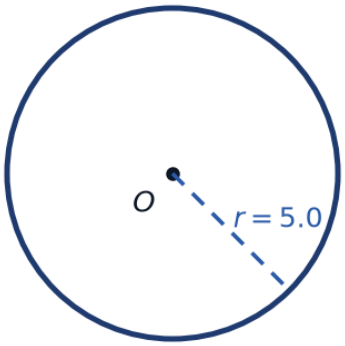
**b.** $7$

**c.** $3$

**d.** $9$

**23** [Foundational] • Target: Volume and surface area

In the coordinate plane, the circle $(x-2)^2+(y-3)^2=25$ intersects the $y$-axis at two points. What is the distance between these two points?




**a.** $2\sqrt{21}$

**b.** $4\sqrt{5}$

**c.** $8$

**d.** $10$

**24** [Foundational] • Target: Similar triangles and proportional reasoning

A sphere has a volume of $288\pi$ cubic centimeters. What is its surface area in square centimeters?

**a.** $144\pi$

**b.** $72\pi$

**c.** $288\pi$

**d.** $216\pi$

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**25** [Foundational] • Target: Angle relationships

What is the area of a circle whose circumference is equal to the perimeter of a square with side length 6?

**a.** $\frac{72}{\pi}$

**b.** $\frac{144}{\pi}$

**c.** $36\pi$

**d.** $144\pi$

**26** [Foundational] • Target: Coordinate geometry: distance and midpoint

In the coordinate plane, the circle $x^2 + y^2 = 25$ intersects the line $y = 3$ at two points. What is the distance between these two points?

**a.** 6

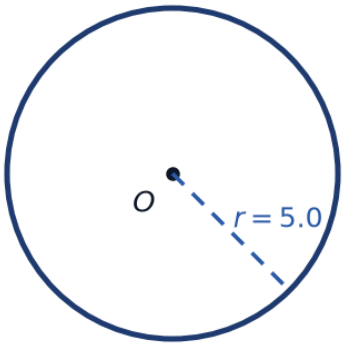
**b.** 8

**c.** 10

**d.** 12

**27** [Foundational] • Target: Circles: area, circumference, and equation

A circle has circumference $10\pi$. What is its area?




**a.** $5\pi$

**b.** $10\pi$

**c.** $25\pi$

**d.** $100\pi$

**28** [Foundational] • Target: Composite and shaded regions

In a circle, a central angle of $60^\circ$ intercepts what fraction of the entire circumference?

**a.** $\frac{1}{4}$

**b.** $\frac{1}{5}$

**c.** $\frac{1}{6}$

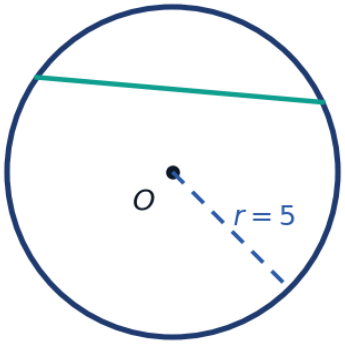
**d.** $\frac{1}{8}$

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: A well-designed word problem: designing a park**

The figure shows a circle with center O. The radius of the circle is 5 units. If the length of chord AB is 8 units, what is the distance from the center O to the chord AB?




**a.** 3 units

**b.** 4 units

**c.** 5 units

**d.** 6 units

**30** **[Intermediate]** • **Target: Area and perimeter**

Triangle $ABC$ is similar to triangle $DEF$, where each side of $DEF$ is 4 times the length of the corresponding side of $ABC$. If the area of triangle $ABC$ is 26 square units, what is the area of triangle $DEF$?

**a.** 104

**b.** 1664

**c.** 436

**d.** 416

**31** **[Intermediate]** • **Target: The Pythagorean theorem**

A circle has a radius of 18 cm. What is the area, in square centimeters, of a sector bounded by a central angle of $30^\circ$?

**a.** $27\pi$

**b.** $54\pi$

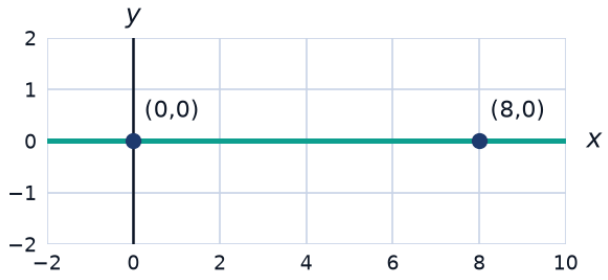
**c.** $13\pi$

**d.** $3\pi$

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**32** **[Intermediate]** • **Target: Volume and surface area**

A circle in the coordinate plane passes through $(0, 0)$, $(8, 0)$, and $(0, 6)$. What is the radius of the circle?




**a.** 4

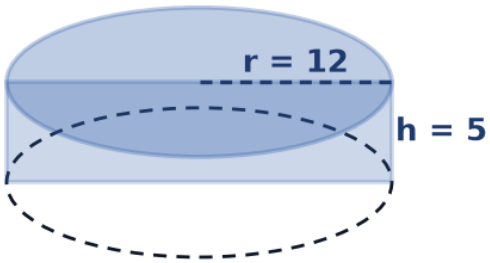
**b.** 5

**c.** 6

**d.** 10

**33** **[Intermediate]** • **Target: Similar triangles and proportional reasoning**

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?




**a.** $100\pi$

**b.** $\frac{400\pi}{3}$

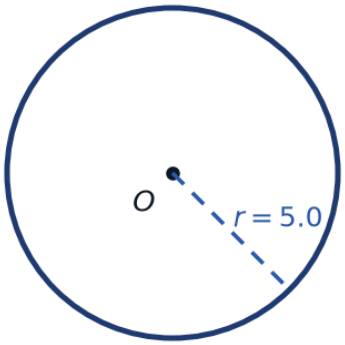
**c.** $\frac{500\pi}{3}$

**d.** $200\pi$

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**34** **[Intermediate]** • **Target: Angle relationships**

The line $x = 5$ intersects the circle $(x - 2)^2 + (y + 1)^2 = 25$ at two points. What is the distance between these two points?




**a.** 6

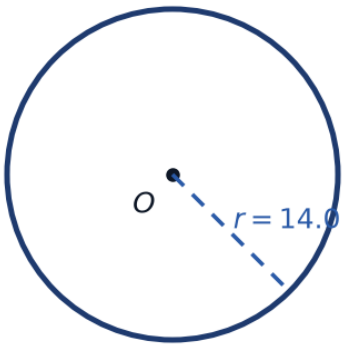
**b.** 8

**c.** 10

**d.** 12

**35** **[Intermediate]** • **Target: Coordinate geometry: distance and midpoint**

What is the circumference of a circle with a diameter of 14 inches?




**a.** $7\pi$ in

**b.** $14\pi$ in

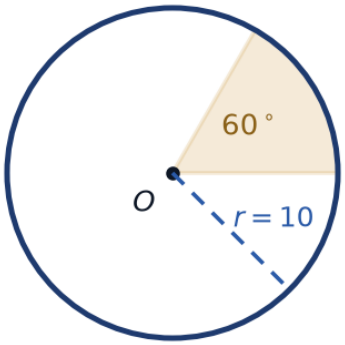
**c.** $28\pi$ in

**d.** $49\pi$ in

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**36** [Intermediate] • Target: Circles: area, circumference, and equation

A chord in a circle of radius 10 cm is at a distance of 6 cm from the center. What is the length of the chord?




**a.** 8 cm

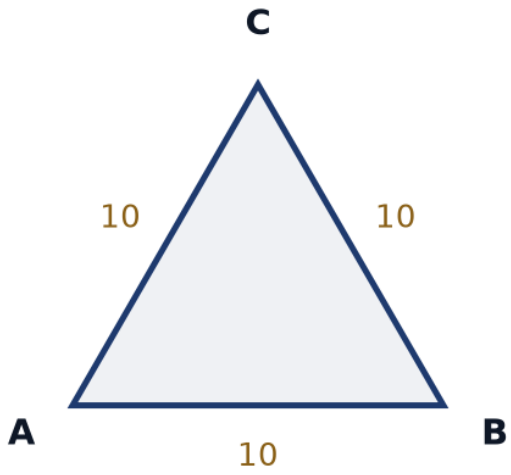
**b.** 12 cm

**c.** 16 cm

**d.** 18 cm

**37** [Intermediate] • Target: Composite and shaded regions

An equilateral triangle has sides of length 10. What is its perimeter?




**a.** 20

**b.** 25

**c.** 30

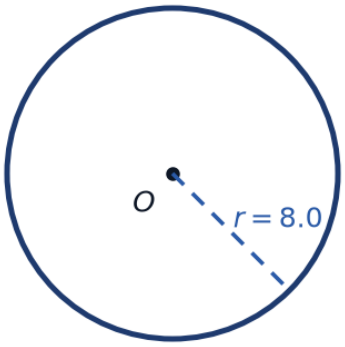
**d.** 35

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** **[Challenging]** • **Target: A well-designed word problem: designing a park** • **Cross-Lesson: 2.01 Quadratics & Polynomials**

A circle with radius 8 has a sector with area $16\pi$. What is the measure of the central angle of this sector in radians?




**a.** $\frac{\pi}{4}$

**b.** $\frac{\pi}{2}$

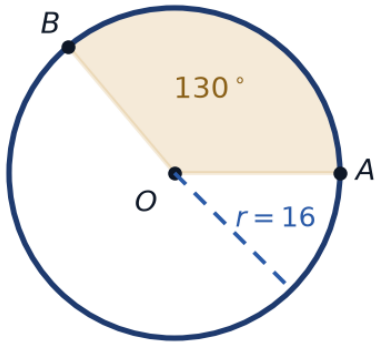
**c.** $\frac{2\pi}{3}$

**d.** $\pi$

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**39** [Challenging] • Target: Area and perimeter • Cross-Lesson: 3.01 Functions & Graphs

A circle has equation $(x - 2)^2 + (y + 1)^2 = 16$. What is the radius?




**a.** 2

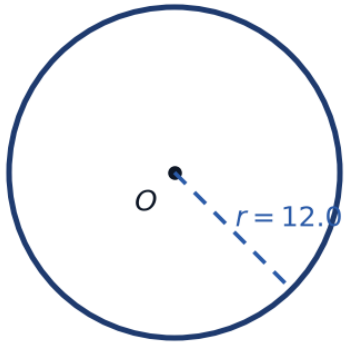
**b.** 4

**c.** 8

**d.** 16

**40** [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 4.01 Ratios, Rates & Data

A circle has radius 12 inches. What is the area of a sector with a central angle of $60^\circ$?




**a.** $12\pi$ in$^2$

**b.** $18\pi$ in$^2$

**c.** $24\pi$ in$^2$

**d.** $36\pi$ in$^2$

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**41** [Challenging] • Target: Volume and surface area • Cross-Lesson: 5.01 Statistics & Probability

A sector of a circle with radius 12 cm has an area of $48\pi$ square centimeters. What is the central angle of the sector in degrees?

**a.** $120^\circ$

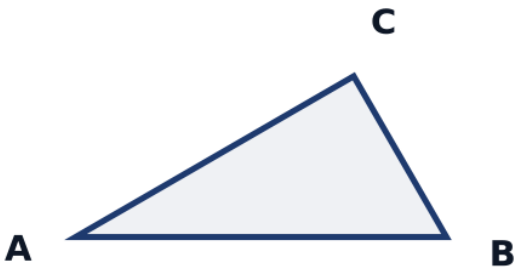
**b.** $60^\circ$

**c.** $90^\circ$

**d.** $150^\circ$

**42** [Challenging] • Target: Similar triangles and proportional reasoning • Cross-Lesson: 7.01 Trigonometry & Additional Topics

In a $30^\circ - 60^\circ - 90^\circ$ triangle, the side opposite the $30^\circ$ angle is 7. What is the length of the side opposite the $60^\circ$ angle?




**a.** $7\sqrt{2}$

**b.** $7\sqrt{3}$

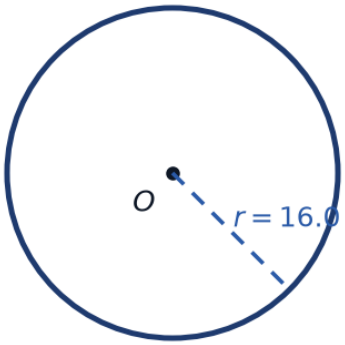
**c.** 14

**d.** $14\sqrt{3}$

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**43** [Challenging] • Target: Angle relationships • Cross-Lesson: 8.01 Exponentials & Logarithms

A central angle of $45^\circ$ in a circle of radius 16 cm intercepts an arc. What is the length of this arc?




**a.** $2\pi$ cm

**b.** $4\pi$ cm

**c.** $6\pi$ cm

**d.** $8\pi$ cm

**44** [Challenging] • Target: Coordinate geometry: distance and midpoint • Cross-Lesson: 1.01 Heart of Algebra

A cylinder has a radius of 4 cm and a height of 9 cm. What is its total surface area in terms of $\pi$?

**a.** $104\pi$

**b.** $72\pi$

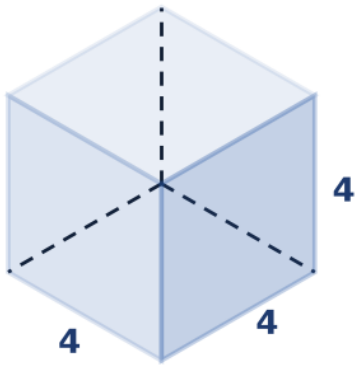
**c.** $144\pi$

**d.** $96\pi$

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**45** [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the surface area of a cube with edge length 4 cm?




**a.** 64 cm²

**b.** 96 cm²

**c.** 128 cm²

**d.** 144 cm²

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